



Molecular and morphological evidence support a new species of Asiatic short-tailed shrew (Eulipotyphla: Soricidae)

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Version of Record, first published online October 13, 2023, with fixed content and layout in compliance with Art. 8.1.3.2 ICZN.

Asiatic short-tailed shrews (*Blarinella* and *Parablarinella*) are small insectivorous mammals distributed mainly in central and southwestern China, and adjacent Myanmar and Vietnam, currently comprising three species. In 2021, we collected five specimens of Asiatic short-tailed shrews in the Dabie Mountains, eastern China, which is geographically distant from the known distribution of the taxa. Herein, we integrated molecular and morphometric approaches to assess the taxonomy, phylogeny, and divergence of Asiatic short-tailed shrews. Our results indicate that the specimens collected from Dabie Mountains represent a new species, formally described here. The new species is morphologically distinguishable from other Asiatic short-tailed shrews by the combination of a triangular upper P⁴, lighter feet color, relatively broader skull, and more curved outline of tooth row of P⁴–M³. Phylogenetic analyses indicate the new species is sister to *P. griselda*, and the *p*-distance of *Cyt b* between the two species is 8.3%. The divergence between the new species and *P. griselda* occurred ca. 3.08 million years ago and may have been affected by East Asia's topographic and climate changes in the late Pliocene.

Key words: *Blarinella*, Dabie Mountains, eastern China, *Parablarinella*, small mammals, taxonomy

Nomenclatural statement.—A life science identifier (LSID) number was obtained for this publication: urn:lsid:zoobank.org:pub:1B9FE96C-8D7A-47B8-B41C-9FEE2E3AF322

Asiatic short-tailed shrews in the genera *Blarinella* and *Parablarinella* are a group of small insectivorous mammals with a stout body, short tail, and intensively dark-pigmented teeth. These animals are mainly distributed in central and southwestern China and adjacent areas (Burgin and He 2018). The Asiatic short-tailed shrews currently comprise the two genera *Blarinella* Thomas, 1911 (suggested common name: West Asian short-tailed shrews) and *Parablarinella* Bannikova et al., 2019 (suggested common name: East Asian short-tailed

shrews)—and three species: *B. quadraticauda* (Milne Edwards, 1872); *B. wardi* Thomas, 1915; and *P. griselda* (Thomas, 1912). A well-resolved taxonomic account of the Asiatic short-tailed shrews is important for biomedical applications and human safety, as a recent study showed that *B. quadraticauda* has oral venom systems, with the venom capable of inducing blood hypercoagulation and hypertension (Liao et al. 2022).

The taxonomy of Asiatic short-tailed shrews has been revised several times. Milne Edwards (1872) described the first

species, *Sorex quadraticauda* (= *Blarinella quadraticauda*) based on a specimen collected from Moupin (= Baoxing) in Sichuan, China. Thomas (1911) found that the species was more closely related to the North American genera *Blarina* and *Cryptotis* than to *Sorex*, consequently establishing the genus *Blarinella* to distinguish it. Tomas described two other species: *B. griselda* based on specimens collected from Taozhou (= Lintan), Gansu, China (Thomas 1912); and *B. wardi* from Hpimaw (= Pianma), Yunnan, China (Thomas 1915). The two species were later considered subspecies of *B. quadraticauda* (Allen 1938; Ellerman and Morrison-Scott 1951; Hoffmann 1987; Corbet and Hill 1992). Based on multivariate analyses of skulls, Jiang et al. (2003) suggested that all three species were valid, a resolution that was accepted subsequently (Hutterer 2005a).

Recent phylogenetic analyses by Chen et al. (2012) supported *B. wardi* as an independent lineage, while *B. quadraticauda* was embedded within *B. griselda*. More recently, He et al. (2018) found that the species previously identified as *B. griselda* formed two deeply divergent lineages, one closer to *B. quadraticauda* and the other comprised of topotypes of *B. griselda*. They, therefore, recognized *B. griselda* under a new genus, *Pantherina*. Bannikova et al. (2019) pointed out that the new genus name previously was assigned to a subgenus of African beetles (genus *Agrilus*) and proposed a new genus name, *Parablarinella*. Currently, *P. griselda* is only recorded in southern Gansu, north Sichuan, southern Shaanxi, and southern Ningxia (Bannikova et al. 2019; Pu et al. 2020); *B. wardi* is distributed in northern Myanmar and western Yunnan, China; and *B. quadraticauda* is widespread in central and southwestern China (Chongqing, Hubei, Sichuan, Shaanxi, Yunnan, Chongqing, Hubei, and Sichuan Provinces) and adjacent North Vietnam. Thus, Western Hubei Province (Changyang County) is currently considered the easternmost distribution of the Asiatic short-tailed shrews (Burgin and He 2018; Bannikova et al. 2019).

During a recent mammal survey in September–October 2021, we collected five specimens of Asiatic short-tailed shrews in the Dabie Mountains, Anhui Province, eastern China. The specimens were interesting because the Asiatic short-tailed shrews were previously thought to be restricted to central and western China (Burgin and He 2018; Bannikova et al. 2019). Our molecular analysis of the mitochondrial cytochrome b (*Cyt b*) gene revealed that these specimens were most closely related to *Parablarinella griselda*, but with a notably high genetic distance, suggesting that they represented an undescribed species (hereafter, *Parablarinella* sp.).

We obtained morphological and molecular data from these specimens and the other three known species of *Blarinella* and *Parablarinella*. Using these data, we assessed the taxonomy, phylogeny, and diversification of the Anhui specimens.

MATERIALS AND METHODS

Five specimens of Asiatic short-tailed shrews were collected in September–October 2021 from Yaoluoping National Nature Reserve, Dabie Mountains, Anhui Province, eastern China

(Fig. 1). Muscle or liver tissue was extracted from all five specimens and stored at -20°C in pure ethanol for molecular studies. The specimens and tissues were deposited at the Biological Museum of Anhui Normal University (AHNU). The collection and handling of animals followed the guidelines of the American Society of Mammalogists (Sikes et al. 2016) and the protocols approved by the Animal Care and Ethics Committee of the AHNU (approval no. AHNU-ET2021002).

Three external measurements—head and body length (HB), tail length (TL), and hindfoot length excluding the claw (HF), were measured in the field to the nearest 1 mm. Nineteen cranial and dental measurements were taken using a digital caliper graduated to 0.01 mm following Chen et al. (2017) and He et al. (2018), including condyloincisive length (CIL), interorbital breadth (IOB), cranial breadth (CB), cranial height (CH), maxillary breadth (MB), rostral length (RL), postrostral length (PRL), palatoincisive length (PIL), postpalatal length (PPL), upper toothrow length (UTR), maximum width across the upper second molars (M^2M^2), distance from the upper fourth premolar to the upper third molar (P^4M^3), mandibular length (ML), lower toothrow length (LTR), length of the lower incisor (LLI), condyle-glenoid length (CGT), breadth of the coronoid process (BCP), postpalatal depth (PPD), and zygomatic plate breadth (ZP).

We also examined specimens of *B. quadraticauda*, *B. wardi*, and *P. griselda* deposited at Kunming Institute of Zoology, Chinese Academy of Sciences (KIZ); American Museum of Natural History (AMNH); National Museum of Natural History, Smithsonian Institution (USNM); and Biological Museum of AHNU. Specimens were identified following He et al. (2018). We obtained 98 specimens assigned to the three recognized species, including 26 *B. wardi*, 67 *B. quadraticauda*, and five *P. griselda* (Appendix I).

Principal component analysis (PCA) with varimax rotation was employed to assess the craniodental variation among species in SPSS 19.0. The cranial and dental measurements were $\log_{10}$ -transformed before the PCA. We further compared the morphology of *Parablarinella* sp. with *P. griselda*, *B. quadraticauda*, and *B. wardi*. The terminologies for morphological descriptions followed Jiang et al. (2003), He et al. (2018), and Bannikova et al. (2019).

The total genomic DNA of each specimen was extracted from muscle or liver tissue using a Qiagen DNeasy blood and tissue kit (Qiagen DNeasy Blood and Tissue Kit, China). We amplified and sequenced two mitochondrial genes (*Cyt b* and *12S rRNA*) and segments of three nuclear genes (apolipoprotein B [*APOB*], breast cancer 1 [*BRCA1*], and recombination-activating protein 2 [*RAG2*]). The primers and annealing temperatures were taken from Chen et al. (2017, 2021). Corresponding sequences of *P. griselda*, *B. quadraticauda*, *B. wardi*, *Episoriculus caudatus* (Horsfield, 1851), and *Crociodura fuliginosa* Blyth, 1855 were downloaded from GenBank (Table 1), with *E. caudatus* and *C. fuliginosa* used as outgroup taxa. All genes were aligned in MUSCLE (Edgar 2004) and further examined manually in MEGA 7 (Kumar et al. 2016). In total, we obtained 4,177-bp-long sequences for each specimen, including 2,044

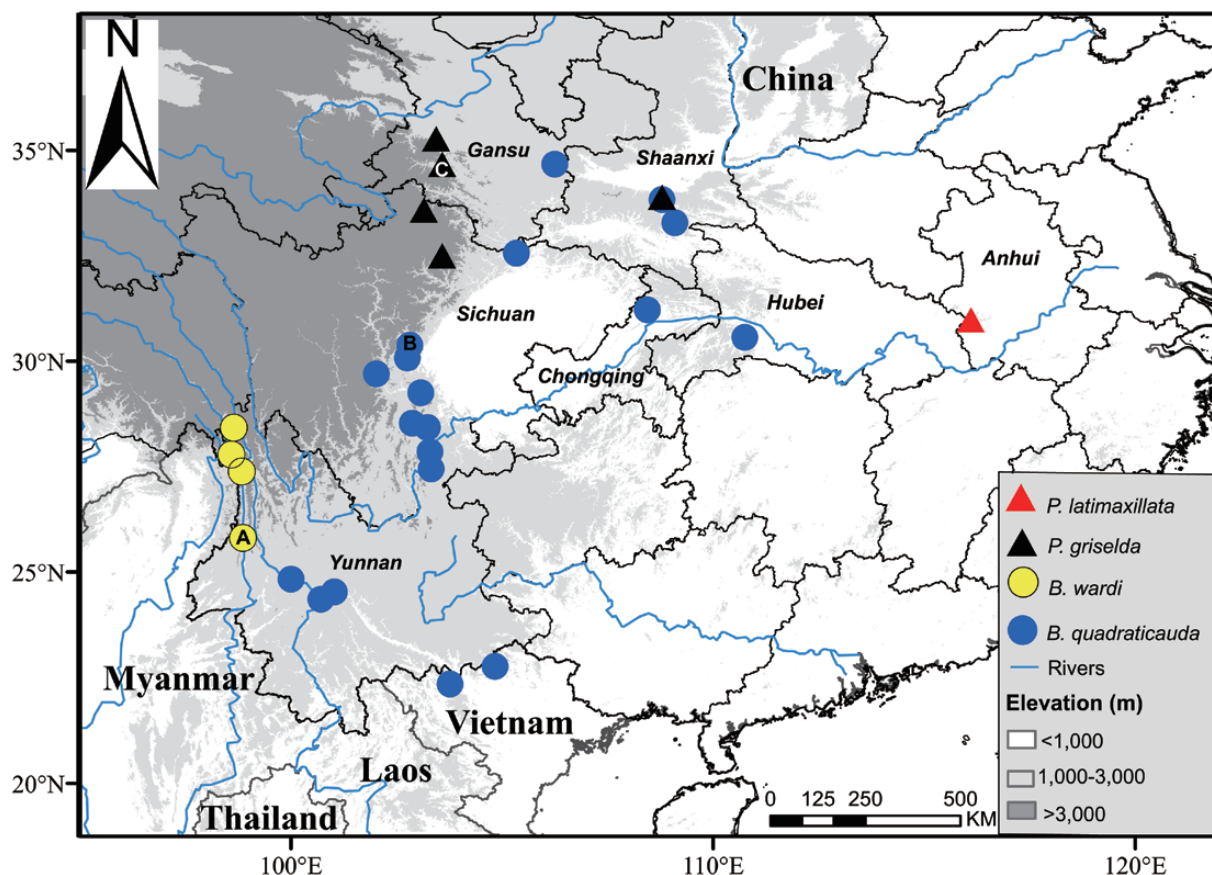


Fig. 1.—Sample localities of specimens used in the present study. (A) Type locality of *Blarinella wardi*, (B) type locality of *B. quadratica*, (C) type locality of *Parablarinella griselda*.

bp of mitochondrial DNA—*Cyt b* (1,140 bp) and *12S rRNA* (904 bp); and 2,133 bp of nuclear DNA—*APOB* (570 bp), *BRCA1* (813 bp), and *RAG2* (750 bp). All new sequences have been deposited in GenBank (accession numbers: OP526371–OP526377 and OP536989–OP537014; Table 1).

We calculated genetic distances (uncorrected *p*-distances) of the complete *Cyt b* gene between *Parablarinella* sp. and *P. griselda*, *B. quadratica*, and *B. wardi* in MEGA 7.

We used maximum likelihood (ML) and Bayesian inference (BI) analyses to reconstruct phylogenetic relationships using IQ-TREE and MrBayes in PhyloSuite (Zhang et al. 2020). The phylogenetic analyses were conducted using: (i) concatenated mitochondrial genes; (ii) each nuclear gene; and (iii) concatenated mitochondrial and nuclear genes. The best-fit partitioning scheme and evolutionary substitution models were estimated using PartitionFinder v.2.0 (Lanfear et al. 2012).

We used BEAST 2.6 (Bouckaert et al. 2014) to estimate divergence times of Asiatic short-tailed shrews based on the concatenated data set of mitochondrial and nuclear genes. We defined data blocks based on genes and codon positions. The best partitioning scheme and substitution models are given in Supplementary Data SD1. We employed the Birth–Death model as tree priors and the relaxed log-normal clock model as the clock model prior. The divergence between Crocidurinae and Soricinae, which is estimated at about 36 Ma (95% confidence interval [CI] = 28.6–44.0 Ma; Springer et al. 2018),

was used as a secondary calibration point. We established the prior using log-normal distribution with a mean of 36 Ma and a standard deviation of 0.135, as per He et al. (2018). Each analysis was run for 100 million generations and sampled every 10,000 generations. The first 10% of the generations were treated as burn-in. Convergence was assessed using TRACER 1.7 (Rambaut et al. 2018) to confirm the effective sample sizes; values greater than 200 were considered adequate.

We conducted Bayesian phylogenetics and phylogeography (BPP) analyses for species delimitations in BPP 3.1 using all five genes based on the A10 and A11 models (Yang and Rannala 2010). To avoid poor mixing during Bayesian analyses, we followed the suggestions of Yang (2015) and performed 12 replicate analyses using the combinations of algorithms 0, algorithms 1, and different priors for models A10 and A11, respectively (Supplementary Data SD2).

RESULTS

External and craniomandibular measurements of each species are provided in Table 2. Similar to the other three species of Asiatic short-tailed shrews, *Parablarinella* sp. has a very short tail—about one-half the head and body length—which is shorter than all other shrews in Soricinae except species in *Soriculus* and *Anourosorex*. Compared with the other three Asiatic short-tailed shrews, *Parablarinella* sp. has relatively

Table 1.—Samples and sequences used in this study. New sequences generated in this study are shown in bold.

Species	Collecting site	Museum code	Abbreviation	<i>Cyt b</i>	<i>12S</i>	<i>APOB</i>	<i>RAG2</i>	<i>BRCA1</i>
<i>Parablarinella</i> sp.	Anqing, Anhui, China	AHNU 202109060	DBS060	OP536999	OP526373	OP536989	OP537006	OP536996
<i>Parablarinella</i> sp.	Anqing, Anhui, China	AHNU 202109061	DBS061	OP537000	OP526374	OP536990	OP537007	OP536997
<i>Parablarinella</i> sp.	Anqing, Anhui, China	AHNU 202109062	DBS062	OP537001	OP526375	OP536991	OP537008	OP536998
<i>Parablarinella</i> sp.	Anqing, Anhui, China	AHNU 202109123	DBS123	OP537002	OP526376	OP536992	OP537009	
<i>Parablarinella</i> sp.	Anqing, Anhui, China	AHNU 202109187	DBS187	OP537003	OP526377	OP536993	OP537010	
<i>Parablarinella griselda</i>	Taizishan, Gansu, China	ZMMU 195179	Pargris1	KY249530		KY249535	KY249547	MN199105
<i>Parablarinella griselda</i>	Ruoergai, Sichuan, China	ZMMU G17-87	Pargris2	MN199121		MN199102	MN199117	
<i>Parablarinella griselda</i>	Songpan, Sichuan, China	ZMMU G18-252	Pargris3	MN199122		MN199103	MN199118	
<i>Parablarinella griselda</i>	Lintan, Gansu, China	NHMUK 1912.8.5.23	Pargris4	MN199123				
<i>Parablarinella griselda</i>	Ningshaan, Shaanxi, China	KIZ 0509390	Pargris5	MH801947		MH801946		
<i>Parablarinella griselda</i>	Ningshaan, Shaanxi, China	KIZ 0509578	Pargris6	MH801948		MH801944		
<i>Parablarinella griselda</i>	Ningshaan, Shaanxi, China	KIZ 0509579	Pargris7	NC_042749	NC_042749	MH801945		
<i>Blarinella quadratica</i>	Sa Pa, Lao Cai, Vietnam	ZIN 96272	Blaquad1			KY249531	KY249542	MN199108
<i>Blarinella quadratica</i>	Sa Pa, Lao Cai, Vietnam	ZIN 97788	Blaquad2	KY249527		KY249532	ZIN 97788	MN199109
<i>Blarinella quadratica</i>	Sa Pa, Lao Cai, Vietnam	ZIN 101574	Blaquad3	KY249528		KY249533	KY249545	MN199107
<i>Blarinella quadratica</i>	Sa Pa, Lao Cai, Vietnam	ZIN 101575	Blaquad4	KY249529		KY249534	KY249546	MN199106
<i>Blarinella quadratica</i>	Sa Pa, Lao Cai, Vietnam	ZMMU G17-12	Blaquad5	MN199120		MN199101	MN199116	MN199110
<i>Blarinella quadratica</i>	Jingdong, Yunnan, China	KIZ 19677	Blaquad6	GU981258	GU981016	GU981108	GU981441	GU981183
<i>Blarinella quadratica</i>	Fengqing, Yunnan, China	KIZ 19702	Blaquad7	GU981259	GU981017	GU981109	GU981442	GU981184
<i>Blarinella wardi</i>	Lushui, Yunnan, China	KIZ 1312906	Blaward1	OP537013	OP526371	OP537011	OP537004	OP536994
<i>Blarinella wardi</i>	Gongshan, Yunnan, China	KIZ 15037	Blaward2	OP537014	OP526372	OP537012	OP537005	OP536995
<i>Blarinella wardi</i>	Gongshan, Yunnan, China	KIZ GS113	Blaward3	MH801937	MH801937	MH801943		
<i>Blarinella wardi</i>	Diqing, Yunnan, China	MLXS 0241	Blaward4 ^a	JF719714		JF719698		
<i>Blarinella wardi</i>	Diqing, Yunnan, China	MLXS 0242	Blaward5 ^a	JF719714		JF719698		
<i>Blarinella wardi</i>	Diqing, Yunnan, China	MLXS 0243	Blaward6 ^a	JF719714		JF719698		
<i>Crocidura fuliginosa</i>	Yunnan, China	KIZ 19701	Crofuli1	GU981271	GU981029	GU981117	GU981450	GU981192
<i>Episoriculus caudatus</i>	Yunnan, China	KIZ 19716	Episcau1	GU981272	GU981030	GU981118	GU981451	GU981193

^aThe three sequences are from one haplotype.

longer head and body length and shorter tail, and averagely larger CB, IOB, MB, and M²M².

The PCA using 19 log₁₀-transformed craniomandibular measurements of intact skulls produced two axes with eigenvalues > 1.0, which counted for 69.5% and 9.4% of the total variation, respectively (Table 3). Loadings on PC1 were positive, indicating that the eigenvector represented overall skull size. PC1 also had the highest loadings for M²M², MB, CB, and IOB suggesting

greater contributions from the width measurements (loading > 0.84; Table 3). The second component (PC2) was strongly and positively correlated with PPL, PRL, CGT, and CIL (loadings > 0.85), indicating it mainly represented skull length. A plot of PC1 and PC2 showed *P. griselda* and *B. quadratica* highly overlapping along both PC1 and PC2. These two species were well-separated from *B. wardi* and *Parablarinella* sp. along PC1 (Fig. 2). *Parablarinella* sp. specimens occupied the lower-right

Table 2.—Mean values (mm), standard deviations, ranges, and sample sizes of *Parablarinella* and *Blarinella* species. Measurements for *P. griselda*, *B. quadraticauda*, and *B. wardi* include data from He et al. (2018).

Variable	<i>Parablarinella</i> sp.	<i>P. griselda</i>	<i>B. quadraticauda</i>	<i>B. wardi</i>
	<i>n</i> = 5	<i>n</i> = 5	<i>n</i> = 66	<i>n</i> = 23
HB	71 ± 3 67–76; 5	67 ± 2 66–70; 4	67 ± 5 52–81; 66	64 ± 3 60–69; 23
TL	36 ± 3 32–39; 5	38 ± 3 35–41; 4	39 ± 5 31–60; 65	38 ± 38 32–43; 21
HF	11 ± 1 11–12; 5	12 ± 2 11–14; 4	12 ± 2 9–16; 65	12 ± 1 11–13; 23
CIL	19.95 ± 0.39 19.42–20.33; 4	19.59 ± 0.37 19.14–19.96; 5	20.36 ± 0.72 19.13–21.69; 66	19.15 ± 0.39 18.54–19.91; 23
IOB	4.80 ± 0.17 4.58–4.97; 4	4.38 ± 0.17 4.20–4.59; 5	4.41 ± 0.19 4.06–4.83; 66	4.04 ± 0.10 3.89–4.33; 23
CB	9.62 ± 0.37 9.09–9.95; 4	9.16 ± 0.32 8.85–9.51; 5	9.24 ± 0.26 8.57–9.70; 65	8.33 ± 0.23 7.84–8.72; 22
CH	5.80 ± 0.05 5.75–5.85; 4	5.53 ± 0.52 4.62–5.89; 5	5.11 ± 0.30 4.43–6.09; 65	4.58 ± 0.21 4.12–4.93; 23
RL	6.87 ± 0.09 6.78–6.98; 4	6.47 ± 0.17 6.32–6.73; 5	7.36 ± 0.38 6.79–8.05; 66	6.64 ± 0.20 6.25–7.11; 23
PRL	11.65 ± 0.28 11.23–11.83; 4	11.60 ± 0.38 11.20–12.04; 5	12.05 ± 0.45 11.26–12.94; 65	11.58 ± 0.30 11.06–12.11; 23
MB	6.39 ± 0.29 6.01–6.71; 4	5.84 ± 0.22 5.57–6.07; 5	5.98 ± 0.18 5.62–6.36; 65	5.39 ± 0.15 5.04–5.77; 23
PIL	9.10 ± 0.19 8.95–9.38; 4	8.66 ± 0.22 8.42–8.90; 5	9.09 ± 0.33 8.26–9.66; 66	8.42 ± 0.23 7.93–8.78; 23
PPL	8.97 ± 0.24 8.64–9.19; 4	9.24 ± 0.37 8.76–9.64; 5	9.49 ± 0.37 8.70–10.25; 65	9.15 ± 0.25 8.67–9.63; 23
UTR	8.89 ± 0.11 8.81–9.05; 4	8.40 ± 0.20 8.16–8.61; 5	8.46 ± 0.29 7.79–9.06; 66	7.63 ± 0.24 6.98–7.97; 23
M ² M ²	5.87 ± 0.13 5.71–6.04; 4	5.26 ± 0.12 5.15–5.44; 5	5.36 ± 0.16 5.08–5.77; 66	4.81 ± 0.11 4.56–5.03; 23
P ² M ³	5.16 ± 0.04 5.14–5.22; 4	4.72 ± 0.15 4.58–4.92; 5	5.09 ± 0.14 4.80–5.41; 66	4.68 ± 0.11 4.37–4.87; 23
ML	10.98 ± 0.27 10.83–11.38; 4	10.32 ± 0.24 10.04–10.61; 5	10.26 ± 0.41 9.39–11.00; 65	9.52 ± 0.30 8.91–9.95; 22
LTR	8.13 ± 0.11 8.06–8.29; 4	7.83 ± 0.16 7.66–8.07; 5	8.24 ± 0.29 7.55–8.77; 65	7.50 ± 0.17 7.08–7.74; 22
LLI	4.51 ± 0.12 4.34–4.62; 4	4.50 ± 0.08 4.40–4.60; 5	4.87 ± 0.29 4.20–5.49; 65	4.39 ± 0.18 4.13–4.79; 22
CGT	8.71 ± 0.26 8.33–8.94; 4	8.74 ± 0.20 8.54–8.96; 5	8.90 ± 0.30 8.36–9.55; 65	8.61 ± 0.20 8.13–8.96; 23
BCP	1.29 ± 0.02 1.26–1.31; 4	1.22 ± 0.08 1.18–1.36; 5	1.24 ± 0.10 1.03–1.45; 65	1.11 ± 0.09 0.98–1.34; 22
PPD	3.60 ± 0.09 3.52–3.70; 4	3.36 ± 0.13 3.23–3.54; 5	3.52 ± 0.13 3.29–3.78; 66	3.21 ± 0.08 3.07–3.34; 23
ZIP	2.31 ± 0.08 2.23–2.41; 4	2.05 ± 0.11 1.91–2.16; 5	2.18 ± 0.12 1.86–2.44; 66	1.97 ± 0.11 1.70–2.20; 23

corner of the plot, indicating that this potentially new species had a broader and shorter skull than the recognized species. *Blarinella wardi* plotted on the negative regions of PC1, suggesting that it had a narrower skull compared to the other species.

The BI and ML phylogenies recovered very similar topologies; therefore, only the ML gene trees are shown (Fig. 3; Supplementary Data SD3). We considered SH-like approximate likelihood-ratio test values (SH-aLRT) ≥ 80 and ultrafast bootstrap supports (UFBoot) ≥ 95 as strong support (Minh et al. 2022). The phylogenetic trees supported the Asiatic short-tailed shrews clustering into two major clades. One clade comprised *B. quadraticauda* and *B. wardi*, and the other comprised *Parablarinella* sp. and *P. griselda*. Sequences of *Parablarinella* sp. formed a monophyletic clade that was strongly supported as sister to *P. griselda* (SH-aLRT ≥ 97 , UFBoot ≥ 99) based

on gene trees for *BRCA1*, *RAG2*, concatenated mitochondrial genes, and concatenated mitochondrial and nuclear genes—however, they formed a distinct clade within *P. griselda* in the *APOB* gene tree (Supplementary Data SD3).

Divergence time estimates showed robust support (Fig. 4) for the genera *Parablarinella* and *Blarinella* that estimated to have diverged ca. 13.42 Ma (95% CI = 9.30–17.83 Ma), while *Parablarinella* sp. and *P. griselda* diverged ca. 3.08 Ma (95% CI = 1.96–4.28 Ma). The pairwise *Cyt b* *p*-distances between *Parablarinella* sp. and *P. griselda*, *B. quadraticauda*, and *B. wardi* were 8.3%, 17.7%, and 18.3%, respectively (Supplementary Data SD4). Under different algorithms, models, and priors, the BPP results consistently supported the four species as distinct with high support (posterior probabilities [PP] ≥ 0.99 ; Supplementary Data SD2).

Table 3.—Factor loadings, eigenvalues, and percentage of variance explained for principal components (PC) 1 and 2.

Variables	PC1	PC2
M ² M ²	0.910	0.264
MB	0.887	0.335
CB	0.846	0.369
IOB	0.844	0.122
UTR	0.777	0.475
ZP	0.750	0.310
PPD	0.728	0.528
ML	0.720	0.491
P ⁴ M ³	0.699	0.583
CH	0.679	0.276
BCP	0.556	0.551
PPL	0.147	0.905
PRL	0.256	0.875
CGT	0.279	0.861
CIL	0.484	0.857
LLI	0.382	0.797
PIL	0.607	0.715
LTR	0.645	0.695
RL	0.494	0.565
Eigenvalue	13.201	1.794
Variance explained	69.481	9.440

Both the morphological and molecular analyses indicated that *Parablarinella* sp. is diagnosable from all other recognized species. According to the diagnosis-and-monophyly-based phylogenetic species concept (Mayden 1997; Gutierrez and Garbino 2018), we recognize it as a new species, which we formally describe below.

TAXONOMIC ACCOUNT

Parablarinella latimaxillata Chen & Jiang species nova

Figs. 5 and 6

Anhui Short-tailed Shrew; 安徽黑齿鼯鼠

Holotype.—AHNU 202109061, an adult male collected by Zhongzheng Chen on 25 September 2021 in China: Anhui Province; Anqing City, Dabie Mountains, Yaoluoping Natural Reserve (30.981 N, 116.085 E [datum: WGS84], 1,100 m). The dried skin, cleaned skull, and carcass preserved in 95% alcohol are deposited at the Biological Museum of AHNU, Wuhu City, Anhui Province, China.

Paratypes.—AHNU 202109060 (♀) and AHNU 202109062(♀), collected by Zhongzheng Chen together with the holotype at Yaoluoping Natural Reserve (30.981 N, 116.085 E [datum: WGS84], 1,100 m); AHNU 202109123(♂) and AHNU 202109187(♀), also collected at Yaoluoping Natural Reserve (30.978 N, 116.115 E [datum: WGS84], 1,665 m), on 26–27 September 2021. These specimens are deposited at the Biological Museum of AHNU.

Etymology.—The specific Latin name of the new species is derived from *lat*—*latus*, meaning “broad,” and *maxilla*—*maxillary*. The adjectival suffix *ata* means “provided with.” The species name refers to the broad maxilla.

Diagnosis.—The pelage is uniformly dark brown (color code follows Smithe (1975)), the tail is about half the head and body length, and the teeth are heavily pigmented, similar to *P. griselda* and *Blarinella* spp. The backs of hands and feet

are much paler than those of *P. griselda* and *Blarinella* spp. The lingual margin of P⁴ is curved, the hypocone is absent, and the tooth is triangular in occlusal view, distinguishing it from *Blarinella* species. The skull is relatively stout and broad, the rostrum abruptly narrows in the premaxillary region, and the outline of the tooth row of P⁴–M³ is curved or rounded, which distinguishes it from *P. griselda* and *Blarinella* species. It has longer UTR and P⁴M³, and broader M²M² and zygomatic plate than *P. griselda*.

Description.—A medium-sized Asiatic short-tailed shrew (HB = 71 ± 3 mm, CIL = 19.95 ± 0.39 mm) in comparison to *P. griselda* (HB = 67 ± 2 mm, CIL = 19.59 ± 0.37 mm), *B. quadraticauda* (HB = 67 ± 5 mm, CIL = 20.36 ± 0.72 mm), and *B. wardi* (HB = 64 ± 3 mm, CIL = 19.15 ± 0.39 mm). The dorsal and ventral pelage are uniformly dark brown (Fig. 5). External morphology of *P. latimaxillata* is similar to *P. griselda* and *Blarinella* spp., with a stout body. Relative to other soricines the pinnae of *P. latimaxillata* are short, being barely visible. The tail is short, about half the head and body length. The tail is covered in short hairs, dark brown above, similar to the dorsal and ventral pelage, and pale below. The backs of hands and feet are light brown, nearly white at the margin. The claws are enlarged, the forefeet are wider than the hindfeet, and the forefeet fingers are longer than those of the hindfeet.

The skull of *P. latimaxillata* is triangular and robust, much heavier than *Sorex*. Relative to other species of *Blarinella* and *Parablarinella*, the rostrum, nasal vacuity, maxillary, and inter-orbital region of *P. latimaxillata* are relatively short and broad (Fig. 6). The braincase is domed and angular, with well-developed ridges along the lateral edges (Fig. 6). The lambdoidal crest is developed and triangular from dorsal view. There are no zygomatic arches, and maxillary arches are stout. From ventral view, the palate is rather broad. The outline of the upper molar region is quadrangular and abruptly tapers in the premolar regions. The outline of the tooth row from P⁴ to M³ is curved or rounded. The dental formula, following the tooth nomenclature of Hutterer (2005b), is I 3/1, C 1/1, P 3/1, M 3/3 (×2) = 32, which is the same as *P. griselda*. The teeth are heavily pigmented dark chestnut or black. The foramen magnum is large and has a ventral notch. The apex of the first incisor is nearly black and extends anteriorly. The angle of the principal (anterior) hooklike cusp to the posterior cusp of I¹ is moderately acute, <90°. There are five upper unicuspid, but only the first four are visible in lateral view. U¹–U³ gradually decreasing in size, are crowded, and slightly lean toward the incisor. U⁴ and U⁵ are minute, less than half the height of U³. The length of U⁴ is slightly less than or equal to the width, and the occlusal surface is triangular or nearly square. U⁵ is nearly circular in occlusal view. The lingual margin of P⁴ is curved, protocone small, hypocone absent, and triangular in occlusal view, similar to *P. griselda*. M¹ and M² are similar, large, and semirectangular, the ectoloph is W-shaped, the protocone is low, no hypocone present, and the posterior edges of these teeth are concave. M³ is reduced, without metastyle and hypocone, and the tooth is only about one-third the size of M².

The mandible is stout, with a short and broad body (Fig. 6). The coronoid process is thick and square and slightly bent

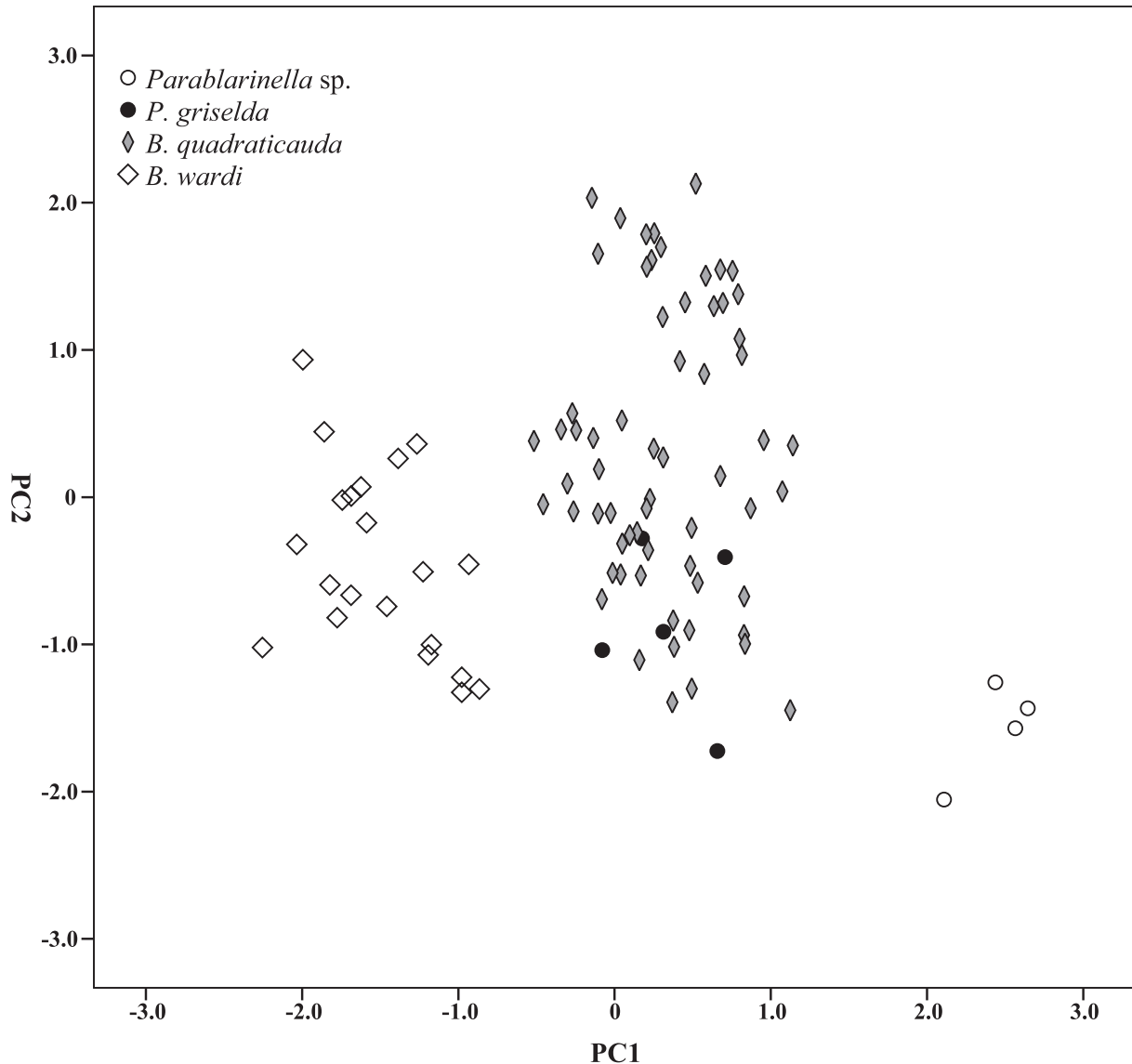


Fig. 2.—Plot of scores on principal components 1 and 2 from analysis of 19 $\log_{10}$ -transformed craniomandibular variables of Asiatic short-tailed shrews.

anteriorly so that the anterior surface is concave. The condyloid process is short and curves toward the braincase, forming an angle less than 90° from the coronoid process. The angular process is well-developed, moderately stout, and straight. The first lower incisor (I_1) is long, with two basal cusps. It is slightly curved upward and has a slender groove on the lingual side. The first lower unicuspid (U_1) is small, sloping forward to the incisor, and the root is squeezed in a narrow space between U_1 and the following premolar (P_4). The P_4 is much bigger than the U_1 , and bulbous-shaped in lingual view. M_1 and M_2 are similar, and “M” shaped in occlusal view. The entoconid is distinct, and the talonid of the two molars is complete in lingual view. M_3 is much smaller than M_1 and M_2 , and the talonid and entoconid structures are reduced.

Comparison.—*Parablarinella latimaxillata* is externally similar to *P. griselda* as well as to the *Blarinella* species (i.e., *B. quadraticauda* and *B. wardi*). The new species can be

distinguished from *P. griselda* by numerous characters. The back of hands and feet of *P. latimaxillata* are light brown and white at the margin, which is paler than those of *P. griselda*. The skull of *P. latimaxillata* is more stout and broader than *P. griselda* (Fig. 6), especially in the maxillary and interorbital regions. Most width-related cranial and dental variables of *P. latimaxillata* are larger than those of *P. griselda* (e.g., M^2M^2 , MB, IOB, and CB; Table 2). In particular, the range of M^2M^2 (5.71–6.04 mm in *P. latimaxillata* vs. 5.15–5.44 mm in *P. griselda*) between the two species does not overlap. The upper molars of *P. latimaxillata* are larger than those of *P. griselda*, so the measurements related to the upper teeth length of *P. latimaxillata* (UTR 8.81–9.05 mm; P^4M^3 5.14–5.22 mm) are much larger than those of *P. griselda* (UTR 8.16–8.61 mm; P^4M^3 4.58–4.92 mm). The zygomatic plate of *P. latimaxillata* also is broader than that of *P. griselda*. The rostrum of *P. latimaxillata* abruptly narrows in the premaxillary region, and the outline of

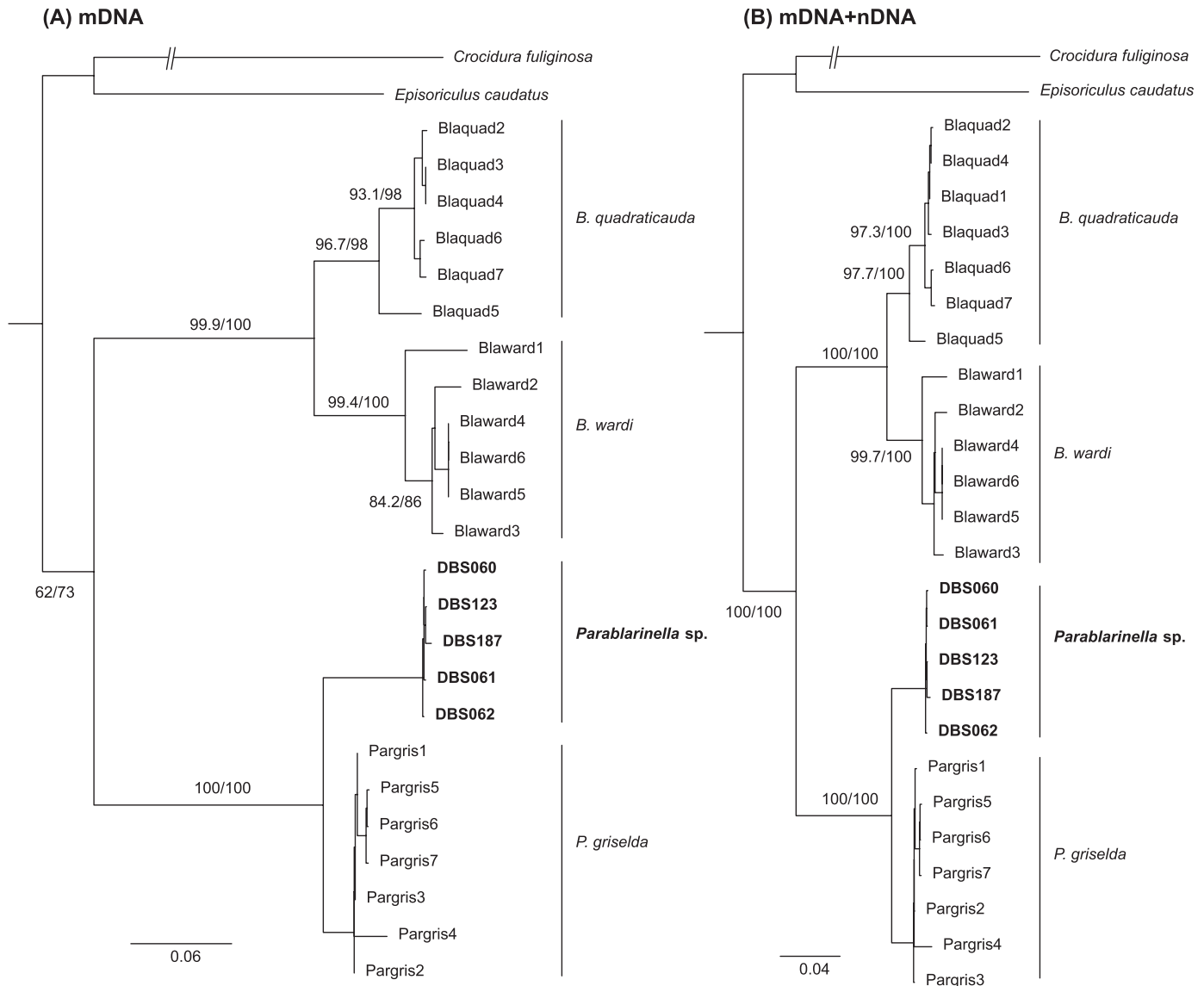


Fig. 3.—Maximum likelihood phylogenetic trees derived from (A) the concatenated mitochondrial gene, and (B) the concatenated mitochondrial-nuclear trees. Branch labels indicate SH-like approximate likelihood ratio test supports (SH-aLRT) and ultrafast bootstrap supports (UFBoot). Scale bars represent substitutions per site.

the tooth row of P^4 – M^3 is curved, forming a shallow semicircle (Fig. 6). In contrast, the rostrum of *P. griselda* tapers gradually in the premaxillary region, and the outlines of the teeth are triangular. The teeth of both species are heavily pigmented; however, the pigmentation of those of *P. latimaxillata* is darker and broader than in *P. griselda* (Fig. 6). The angle of the principal to the posterior cusp of I_1 is moderately acute in *P. latimaxillata*, mostly less than 90° . In contrast, the angle exhibits intraspecific variation in *P. griselda*. Bannikova et al. (2019) pointed out that the angle of the principal to the posterior cusp of I_1 is moderately shallow, greater than 90° in *Parablarinella*, which is diagnosable from *Blarinella*. However, we found these structures exhibit intraspecific variation in *P. griselda*, while the angle is acute and less than 90° in *P. latimaxillata*, indicating it is not a stable character for distinguishing *Parablarinella* from *Blarinella*.

Parablarinella latimaxillata can be distinguished from *Blarinella* spp. by several diagnostic skull and dental characteristics listed by He et al. (2018) and Bannikova et al. (2019): the upper P^4 of *P. latimaxillata* is triangular, and the hypocone is absent (P^4 is trapezoidal, and the hypocone is low but distinct in *Blarinella*); the tip of the coronoid process slightly bends anteriorly (vertically in *Blarinella*), and I_1 is bicuspidate (tricuspidate in *Blarinella*). Furthermore, the light brown hands and feet, broader skull, and the curved or rounded outline of the tooth row of P^4 – M^3 in *P. latimaxillata* also distinguish it from *B. quadraticauda* and *B. wardi*. In the cranial dimensions, most measurements of *P. latimaxillata* (i.e., IOB, CB, CH, MB, PIL, UTR, M^2M^2 , ML, P^4M^3 , PPD, LTR, and ZB) are larger than those of *B. wardi*.

Distribution and habitat.—*Parablarinella latimaxillata* currently only is known from the type locality at Yaoluoping

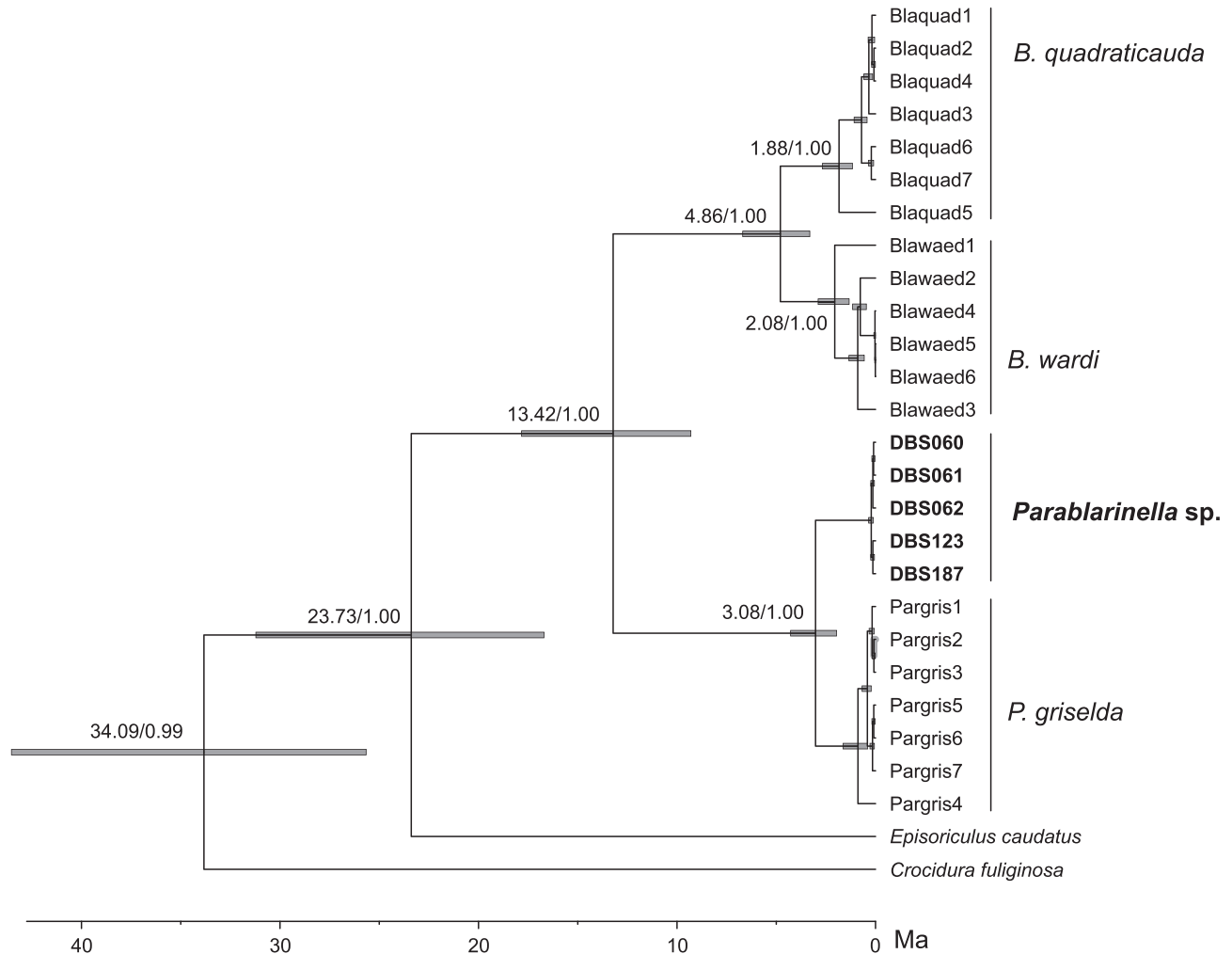


Fig. 4.—Divergence times estimated using BEAST based on mitochondrial-nuclear concatenated data.

natural reserve, Dabie Mountains, Anhui Province, eastern China. All the specimens were collected from broad-leaf deciduous forests at altitudes ranging from 1,100 to 1,700 m.

DISCUSSION

The Asiatic short-tailed shrews were previously thought to be restricted to central and southwestern China and the adjacent Myanmar and Vietnam (He et al. 2018; Bannikova et al. 2019). The distribution of *P. latimaxillata* is geographically distant from the other congeners. All specimens of *P. latimaxillata* available to us were collected from high elevational (i.e., 1,100–1,700 m) alpine forest habitat of the Dabie Mountains (the peak of the mountain is 1,777 m). We did not capture specimens resembling *P. latimaxillata* from our recent extensive surveys in southern Anhui (Pei et al. 2022a, 2022b). The Yangtze River likely acts as a barrier preventing *P. latimaxillata* from migrating southwards. The distribution of *P. latimaxillata* unlikely extends eastward or northward as there are no suitable habitats—the species may have a restricted distribution in Dabie Mountains, far apart from the other Asiatic short-tailed shrews. This discontinuous distribution

pattern is similar to that of two other new species in the Order Eulipotyphla recently discovered in the Dabie Mountains—the shrew-like mole *Uropsilus dabieshanensis* Hu et al., 2021 and the nectogaline shrew *Chodsigoa dabieshanensis* Chen et al., 2022 (Hu et al. 2021; Chen et al. 2022). *Parablarinella latimaxillata* is genetically and morphologically distinctive from all the other Asiatic short-tailed shrews. Such differentiation could have resulted from genetic drift following an ancestral population bottleneck or adaptation to new environments. Notably, *B. quadraticauda* was widely spread along PC2 in Fig. 2, similar to the findings of Jiang et al. (2003); most “*Blarinella griselda*” in that study are now included in *B. quadraticauda*. *Blarinella quadraticauda* has a broad distribution, morphological differentiation (Jiang et al. 2003), and complex genetic structure (Chen et al. 2012; He et al. 2018). A revision of the taxonomy of this species is warranted.

We observed incongruences between the *APOB* gene tree and other gene trees (Supplementary Data SD3). Such incongruence appears to be widespread and may be explained by one of several hypotheses—including recent hybridization and introgression, rapid speciation, and incomplete lineage sorting (He et al. 2010; Chen et al. 2022; Zhang et al. 2022). Under

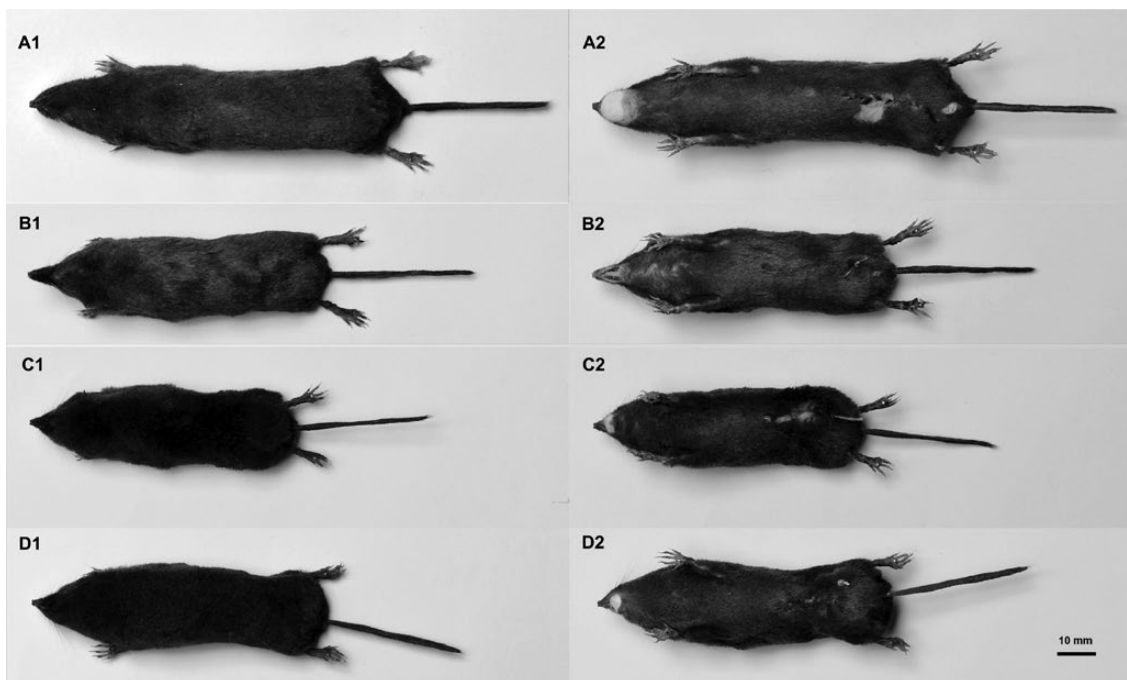


Fig. 5.—Dorsal and ventral views of (A) the holotype of *Parablarinella latimaxillata* species nova (AHNU 202109061), (B) *P. griselda* (KIZ 202109025), (C) *Blarinella quadratica* (KIZ 1311759), and (D) *B. wardi* (KIZ 13121057).

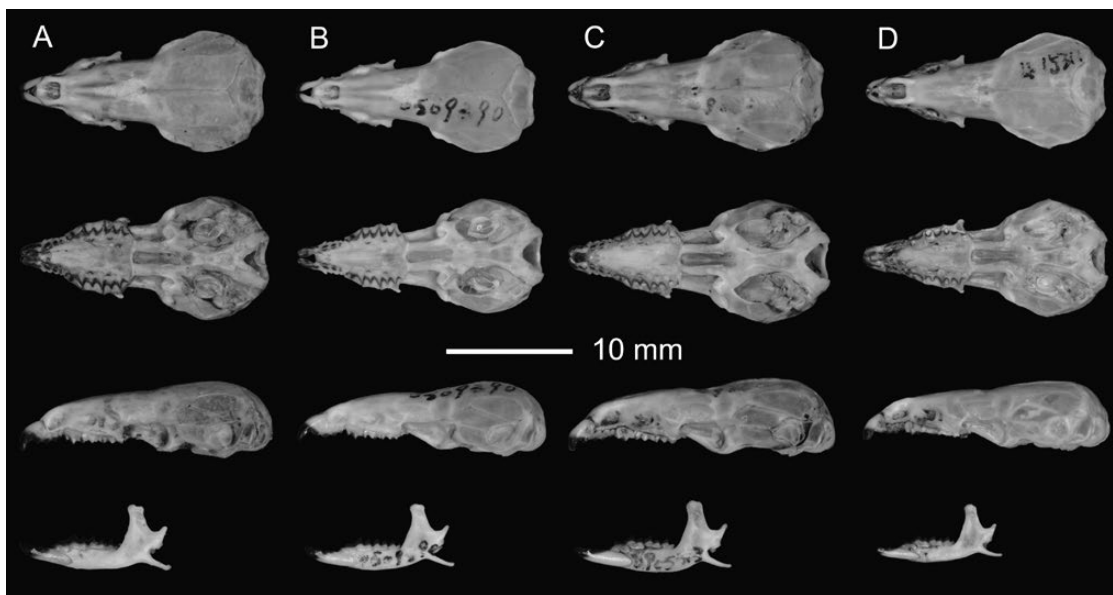


Fig. 6.—Dorsal, ventral, and lateral views of the skull and lateral views of the dentary of (A) the holotype of *Parablarinella latimaxillata* species nova (AHNU 202109061), (B) *P. griselda* (KIZ 0509390), (C) *Blarinella quadratica* (KIZ 0905313), and (D) *B. wardi* (KIZ H10415311).

recent hybridization and introgression, different species would share many sequences (Chen et al. 2017). However, this scenario was not observed in our study. Although *P. latimaxillata* was embedded within *P. griselda* in the *APOB* gene tree, it was still supported as a monophyletic clade. The inconsistent relationship between *P. latimaxillata* and *P. griselda* is more likely due to rapid speciation and incomplete lineage sorting. Both of these hypotheses could lead to informative DNA substitutions that do not have sufficient time to evolve, resulting

in incongruence among gene trees (Jackson et al. 2009). These two hypotheses are not mutually exclusive, and to distinguish them requires more nuclear genes (Townsend 2007).

The divergence of *P. latimaxillata* and *P. griselda* appears to have occurred in the Pliocene period, similar to the divergence between *U. dabieshanensis* and its congeneric species from southwestern and central China (Hu et al. 2021). Geographic and paleontological evidence has suggested that eastern Asia—including the Dabie Mountains—experienced cooling and

aridification during this time (Feng 1976; Qiu and Li 2005; Ge et al. 2013), which triggered the differentiation of mammalian faunas.

There are few fossil records of small mammals from eastern China. Fossils of Asiatic short-tailed shrews were found in late Miocene strata at Lufeng, Yunnan (*Blarinella* sp.), and in late Pleistocene sediments at Wushan, Chongqing (*Blarinella* sp. and *Blarinella quadratica*; Qiu and Storch 2005), but never in eastern China. Our results imply that the Dabie Mountains acted as refugia in the Pliocene and Pleistocene for mountainous species, and the ancestor of *P. latimaxillata* might have migrated southward from Mt. Qinling through a previously unknown corridor. This hypothesis could potentially explain the existence of *U. dabieshanensis*, *C. dabieshanensis*, and the forest hedgehog *Mesechinus hughii* (Thomas 1908; Chen et al. 2020) in eastern China, whose relatives are mainly distributed in the Mt. Qinling region and southwestern China. Assuming the corridor did exist, more currently unknown small mammal species are expected to be discovered in the Dabie, Wuyi, and Nanling mountain areas located in eastern China.

ACKNOWLEDGMENTS

We thank Xiaoxin Pei, Xueyang Ren, Shidong Pu, and Shichun Pu for specimen collection. We thank the Administrative Committee of Yaoluoping National Nature Reserve for their field support.

AUTHOR CONTRIBUTIONS

ZZC and XLJ conceived the ideas and designed methodology; ZZC, JXH, KH, BWZ, YCZ, JC, and KZ collected the samples; JXH generated the sequences; ZZC analyzed the data and led the writing of the manuscript. KOO, KH, and XLJ contributed to editing the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

FUNDING

This research was funded by the National Natural Science Foundation of China (no. 31900318) and Anhui Provincial Natural Science Foundation (2008085QC106), and the University Synergy Innovation Program of Anhui Province (GXXT-2020-075).

CONFLICT OF INTEREST

The authors have no financial or proprietary interests in any material discussed in this article.

DATA AVAILABILITY

All the new sequences obtained in this study have been deposited in GenBank under accession nos. OP526371–OP526377 and OP536989–OP537014.

SUPPLEMENTARY DATA

Supplementary data are available at *Journal of Mammalogy* online.

Supplementary Data SD1.—The best molecular evolutionary model of each partition used in BEAST analyses.

Supplementary Data SD2.—Summarized results of BPP analyses using different algorithms, models, and parameters.

Supplementary Data SD3.—Maximum likelihood phylogenetic trees of the three nuclear genes. Branch labels indicate SH-aLRT and UFBoot support values. Scale bars represent substitutions per site.

Supplementary Data SD4.—The *p*-distances between the Asiatic short-tailed shrews based on the *Cyt b* gene.

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- Submitted 10 August 2022. Accepted 1 September 2023.*
- Associate Editor was Michael McGowen.*
- APPENDIX I**
- Specimens examined for this study. Abbreviations: KIZ, Kunming Institute of Zoology, Chinese Academy of Sciences; AHNU, Anhui Normal University; AMNH, American Museum of Natural History; USNM, National Museum of Natural History, Smithsonian Institution.
- P. latimaxillata* ($n = 5$): Dabie Mountain, Anhui (AHNU 202109060–062, AHNU 202109123, AHNU 202109187).
- P. griselda* ($n = 5$): Gansu (AMNH 60449); Mt. Qinling, Shaanxi (KIZ 0509390, KIZ 0509578–579, KIZ 202109025).
- B. quadraticauda* ($n = 67$): Mt. Qinling, Shaanxi (KIZ 0509420, KIZ 0509423, KIZ 202109086); Baoxin, Sichuan (KIZ 820495, KIZ 820588, KIZ 820625, KIZ 820626); Qionglai, Sichuan (AMNH 111121, AMNH 111125, AMNH 111126–128, AMNH 111130, AMNH 111136); Wenchuan, Sichuan (AMNH 111112, AMNH 111114, AMNH 111119, AMNH 111137, AMNH 111138, AMNH 111140, AMNH 111142, AMNH 111145); Jingdong, Yunnan (KIZ 640068, KIZ 640105, KIZ 640128–129, KIZ 640133–134, KIZ 640152, KIZ 640191, KIZ 640199–200, KIZ 640418, KIZ 640429–432, KIZ 96180, KIZ 98069, KIZ 98107, KIZ 98114, KIZ 98151, KIZ 98152, KIZ 98313, KIZ 98335, KIZ 98378, KIZ 98553, KIZ 72053); Shuangbai, Yunnan (KIZ 1311759); Luchun, Yunnan (KIZ 72053), Nanchuan, Sichuan (KIZ 88126–127, KIZ 88252); Shimian, Sichuan (KIZ 820742–744, KIZ 820752–754, USNM 574288–293); Wulong, Sichuan (KIZ 88320–321, KIZ 88395).
- B. wardi* ($n = 26$): Yunnan, China (KIZ 73863, KIZ 79769, KIZ 810147, KIZ 73889, KIZ 73910, KIZ 73920, KIZ 73927–929, KIZ 73938, KIZ 820280, KIZ 1312906, KIZ 15037, USNM 241403, KIZ 13121057); Myanmar (AMNH 114720–723, AMNH 114739, AMNH 114740–741, AMNH 114748, AMNH 114757–758, AMNH 114760).